

Trigonometry

Identities and equations

YOUR AIM TODAY

Simplify expressions and solve trigonometric equations over an interval.

YOUR RULE

Read the notes and copy each worked step. Attempt every question before opening page 4. Mark in another colour and repair every error.

Student: Rohan

Start time: _____ Finish time: _____

Date: _____

The ideas you need

An identity is true for every allowed angle; an equation is true only for selected angles.

- $\sin^2 x + \cos^2 x = 1$.
- $\tan x = \sin x / \cos x$.
- For $\sin x = k$, use the sine symmetry; for $\cos x = k$, use cosine symmetry.
- For $\tan x = k$, solutions repeat every 180 degrees.
- Always check the stated interval.

MEMORY RULE

Find the reference angle, choose the correct quadrants, then check the interval.

Identity

WORKED STEPS

$$(1 - \cos^2 x) / \sin x = \sin^2 x / \sin x = \sin x.$$

Equation

WORKED STEPS

$$2\sin x = 1 \text{ on } 0 \text{ to } 360: \sin x = 1/2, \text{ so } x = 30, 150 \text{ degrees.}$$

Independent practice

Show every step. Include units in Mechanics.

Question	Working and final answer
1. Simplify $\sin x / \cos x$.	
2. Simplify $1 - \sin^2 x$.	
3. Solve $\sin x = \sqrt{3}/2$ for $0 \leq x \leq 360$ degrees.	
4. Solve $\cos x = -1/2$ on the same interval.	
5. Solve $\tan x = 1$ on the same interval.	
6. Solve $2\cos^2 x - 1 = 0$ on the same interval.	

STOP

Do not continue until all six questions have an attempt.

Mark, diagnose, correct

A wrong answer is useful only when you repair the first wrong step.

Q	Answer and essential working
1	$\tan x$.
2	$\cos^2 x$.
3	60, 120 degrees.
4	120, 240 degrees.
5	45, 225 degrees.
6	$\cos^2 x = 1/2$, so $x = 45, 135, 225, 315$ degrees.

Q	First wrong step	Correct rule

SEND FOR MARKING

Send pages 3 and 4. Ask for a score, the first error in each answer, and one follow-up question per weak skill.